

JOSHUA KARTHIK ASHOK

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Aspiring Backend & AI Systems Engineer passionate about building scalable applications using advanced backend architectures, cloud technologies, and SAP ABAP for enterprise-grade workflows.

Education

Amrita Vishwa Vidyapeetham Aug 2023 – Present
B.Tech in Computer and Engineering Coimbatore, TN
◦ CGPA: 7.61 (Expected date of graduation: July 2027)

Work Experience

TrustyBytes
Full Stack Developer Intern
◦ Engineered **React frontends** and **Node.js/MongoDB APIs**, achieving **20–30% faster UI load times** and **25% improved API reliability** through optimized components, JWT auth hardening, and structured data validation.
◦ Strengthened product quality by adding **70% test coverage** for new features and maintaining CI pipelines, reducing deployment issues by **40%** while contributing to Agile sprint planning and code reviews.

Hyundai
SAP ABAP Intern
◦ Engineered and enhanced **SAP ABAP artifacts** — including **SmartForms**, **DDIC objects**, **BAPIs**, **BADIs**, and **user-exits** — streamlining core manufacturing and procurement workflows and enabling **30% reduction in manual processing overhead** across SAP modules (MM, SD).
◦ Optimized legacy ABAP programs through **performance tuning**, **SQL trace analysis**, and **buffering strategies**, improving **runtime efficiency by 20–25%** and elevating system reliability in production-grade environments.

Projects

ClarityLegal [Live](#) [GitHub](#)
AI-Driven Legal Document Intelligence Platform
◦ Engineered an AI-powered legal analysis system using **Vertex AI (Gemini)**, **OCR**, **RAG search**, **FastAPI**, and **Next.js**, enabling **<5s document processing** and structured extraction of summaries, risks, and key clauses.
◦ Developed **clause extraction**, **risk scoring**, and **grounded Q&A** modules with near-zero hallucination by constraining answers to document-bound evidence and enforcing source attribution.
◦ Implemented semantic search over multi-document embeddings for real-time cross-document snippet retrieval; validated on SaaS, NDA, and employment contracts and presented as a commercial prototype for SMEs.

OptiWare [Live](#) [GitHub](#)
Edge-AI Warehouse Automation
◦ Developed an edge-AI warehouse automation system using **Sony IMX-500**, **Raspberry Pi**, and **TensorFlow Lite**, achieving **sub-200 ms inference latency** with high-precision item detection for real-time sorting.
◦ Integrated **PostgreSQL** and **AWS (SNS/S3/Lambda)** to enable **<1.5s event propagation**, automated alerting, and operator dashboards for end-to-end inventory visibility.
◦ Optimized the edge processing pipeline to reduce overall computational overhead by **30%**, ensuring stable low-latency performance across the full proof-of-concept deployment.

modernERP [Live](#) [GitHub](#)
Cloud ERP with AI & ESG Analytics
◦ Architected an AI-assisted ERP framework for MSMEs targeting **25–40% reduction in procurement cycle time** through demand forecasting, automated purchase workflows, and intelligent approval routing.
◦ Designed a modular cloud architecture with **blockchain-backed audit trails** and **ESG analytics**, enabling **immutable transaction logging** and **real-time sustainability scoring** across supply-chain events.
◦ Delivered a full technical-commercial blueprint, demonstrating **SaaS monetization potential**, **50% reduction in compliance overhead**, and operational gains validated through scenario modelling for MSME workflows.

Publications

Navigating the Dual Realities of Image Generation and Deepfake Detection [Research Paper](#)
◦ Conducted a comparative study of GAN- and diffusion-based image generation models, identified key robustness gaps in deepfake detection (adversarial vulnerability, compression artefacts, cross-model inconsistency), and proposed methodological directions—spanning multimodal cues and feature-level analysis—to improve forensic reliability.

Achievements & Certifications

SEED Business Challenge 2026 (University of Maryland) — Winner [LinkedIn](#)
Google Gen AI Exchange Hackathon — Top 100 / 30,000+ (Project: ClarityLegal) [LinkedIn](#)
Sony AITRIOS Hackathon — Top 10 / 10,000+ (Project: OptiWare) [LinkedIn](#)
Odoo Hackathon — Top 10 / 1,500 (Full-stack rental system) [Google Drive](#)
SIH 2024 — Selected to Regionals (Project: modernERP) [LinkedIn](#)
IITM Foundational Degree — Foundational in Data Science [View Certificate](#)

Technical Skills

Languages: Python, TypeScript/JavaScript, C/C++ (familiar: Java, SAP ABAP)
Backend : FastAPI, Node.js (Express), Next.js API Routes
Databases: PostgreSQL, MongoDB, Firestore, Redis (basic)
AI/Computer Vision : OpenCV, TensorFlow, OCR pipelines
Cloud/DevOps : Docker, Git, CI/CD, Vercel (hands-on: GCP Cloud Run)